

GNN-SDM: Species Distribution Modelling with Graph Neural Networks for Swiss Flora at 30 m Resolution

Marco Kronenberg
`marco.kronenberg@bafu.admin.ch`

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Abstract

Traditional species distribution models treat occurrences as independent points, ignoring the spatial structure of landscapes that governs habitat connectivity and species dispersal. We adapt the GNN-SDM framework (Wu et al., 2025) to Swiss vascular plants at 30 m resolution, constructing a landscape graph of 166,464 patches linked by 465,833 adjacency edges. Using a GraphSAGE architecture tuned for flora, we model habitat suitability for 3,751 species and compare against a Random Forest baseline. [Results summary to be added.]

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1 Introduction

Biodiversity is declining at an unprecedented rate worldwide, driven by habitat loss, fragmentation, climate change, and land-use intensification (IPBES, 2019). Switzerland, despite its relatively small area, harbours exceptional plant diversity owing to its pronounced altitudinal gradients and mosaic of biogeographic regions. Yet even here, a third of all assessed species and half of all habitat types are threatened (Swiss Federal Council, 2022), with 28% of vascular plants considered endangered or regionally extinct and an additional 14% near-threatened (InfoFlora2024). The main drivers—soil sealing, landscape fragmentation, intensive land use, and nitrogen inputs—continue to erode habitat quality and connectivity (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Effective conservation planning requires spatially explicit predictions of where species can persist, and crucially, which landscape elements link suitable habitat patches into functional networks.

Species distribution models (SDMs) have become a cornerstone of conservation biogeography, relating georeferenced occurrence records to environmental predictors to map habitat suitability across space (Elith and Leathwick, 2009; Guisan and Thuiller, 2005). Classical approaches, such as generalised linear models, maximum entropy (Phillips et al., 2006), and ensemble methods such as Random Forests (Breiman, 2001), treat each occurrence as an independent sample drawn from an environmental niche. While powerful, this point-based paradigm ignores the spatial structure of landscapes: two patches with identical environmental conditions may differ dramatically in ecological value depending on their size, shape, and connectivity to neighbouring habitat. For species whose long-term persistence depends on seed dispersal between habitat patches, such as the *Cypripedium calceolus*, this omission can lead to misleading suitability estimates and suboptimal conservation recommendations.

Graph Neural Networks (GNNs) offer a principled way to incorporate landscape structure into SDMs. By representing the landscape as a graph, where nodes are ecologically homogeneous patches and edges encode spatial adjacency. A GNN can learn how environmental conditions in neighbouring patches jointly influence local habitat suitability. Wu et al. (2025) introduced the SOM-GNN-SDM framework, demonstrating that a GraphSAGE architecture (Hamilton et al., 2017) trained on landscape graphs substantially outperforms traditional SDMs for virtual and real species across continental extents. Their approach constructs landscape patches via Self-Organising Maps (SOMs) (Kohonen, 1982), builds an adjacency graph, and propagates information across the graph to produce suitability predictions that are spatially coherent and connectivity-aware.

Beyond improving predictive accuracy, the graph representation opens a direct path to conservation applications. The Swiss federal government has identified the lack of an ecological infrastructure that protects and connects core biodiversity areas as a key deficit, setting a target of 17% of national territory as connected core areas by 2030, up from 13.4% today (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Because the landscape graph explicitly encodes connectivity, it can be analysed using circuit theory and centrality metrics to identify critical corridors: patches whose protection would maintain or restore habitat connectivity for threatened species. This transforms the SDM output from a static suitability map into an actionable tool for spatial conservation prioritisation and gap analysis, directly informing where the ecological infrastructure should be expanded.

In this work, we adapt the GNN-SDM framework to Swiss vascular flora at 30 m resolution, presenting a substantially finer grain than the original continental-scale application. We construct a landscape graph of 166,464 patches from 12 environmental features using SOM-based clustering and connected-component labelling, train per-species GraphSAGE models for 3,751 qualifying plant species, and benchmark against a Random Forest baseline. We further apply circuit-theory connectivity analysis to the resulting suitability maps to identify conservation priority areas and quantify protection gaps relative to the existing protected-area network.

Specifically, we address the following research questions:

1. Can GNN-SDM outperform a traditional Random Forest at patch-level species distribution modelling for Swiss plants?
2. Does the graph structure confer greater benefit for rare and vulnerable species, whose distributions are more fragmented?
3. What GNN architecture is best suited for flora—given that plants, unlike the mobile fauna studied by Wu et al. (2025), have limited active dispersal?
4. Can the graph structure be leveraged to identify spatial gaps in the protected-area network where additional conservation measures would be most beneficial?

2 Data and Study Area

2.1 Study Area

The study area encompasses the entirety of Switzerland and a surrounding buffer zone, defined by the spatial extent of the CHELSA high-resolution climate dataset (WSL, 2024). The bounding box spans from 45.57° N to 48.07° N latitude and 5.72° E to 10.66° E longitude, covering approximately 41,400 km² of land area. This extent includes a border buffer of roughly 10 km into neighbouring countries to avoid edge effects in spatial analyses such as flow accumulation and neighbourhood-based feature computation.

All environmental layers are aligned to a common 30 m master grid derived from the Copernicus GLO-30 Digital Elevation Model (European Space Agency, 2023), projected in geographic coordinates (EPSG:4326). Switzerland’s topography ranges from 193 m (Lake Maggiore) to 4,634 m (Dufourspitze), creating steep environmental gradients over short distances—a property that motivates the use of high-resolution modelling.

2.2 Environmental Features

We compiled 21 candidate environmental features from four thematic groups (Table 1): terrain, land cover, climate, and biological/anthropogenic variables. All layers were processed to the 30 m master grid using source-appropriate resampling methods.

Terrain features. Elevation was obtained directly from the Copernicus GLO-30 DEM (European Space Agency, 2023). Slope and aspect were derived using first-order finite differences at 30 m resolution. Aspect was encoded as two variables (sin and cos of the azimuth angle) to avoid the circular discontinuity at 0°/360°. The Topographic Wetness Index (TWI) was computed using D8 flow direction and accumulation via the `pysheds` library, with pit-filling and depression resolution applied beforehand. Profile and plan curvature were derived from second-order finite differences and smoothed with a 5×5 mean filter to reduce noise.

Land cover. Six land-cover fraction layers were computed from ESA WorldCover 2021 (European Space Agency, 2022) at its native 10 m resolution. For each class, a binary mask was reprojected to the 30 m grid using average resampling, yielding the fraction of each 30 m pixel covered by that land-cover type. The six classes retained for Switzerland are: tree cover, grassland, cropland, built-up, snow/ice, and permanent water bodies.

Climate. Five bioclimatic variables were derived from CHELSA monthly temperature and precipitation normals (1981–2010) at 25 m resolution (WSL, 2024): annual mean temperature (Bio1), temperature seasonality (Bio4, standard deviation), annual precipitation (Bio12), precipitation seasonality (Bio15, coefficient of variation), and precipitation of the warmest quarter (Bio18). Each variable was regridded to the 30 m master grid using bilinear interpolation.

Table 1: Environmental predictor variables compiled for the study. All layers are aligned to the 30 m master grid. The 12 features selected for modelling (after feature selection, Section 2.5) are marked with ●.

Group	Feature	Source	Native res.	Sel.
Terrain	Elevation	Copernicus GLO-30 DEM	30 m	●
	Slope	Derived from DEM	30 m	●
	Aspect (sin, cos)	Derived from DEM	30 m	●
	Topographic Wetness Index	D8 flow accumulation	30 m	●
	Profile curvature	2 nd -order finite diff.	30 m	
	Plan curvature	2 nd -order finite diff.	30 m	
Land cover	Tree cover fraction	ESA WorldCover 2021	10 m	●
	Grassland fraction	ESA WorldCover 2021	10 m	●
	Cropland fraction	ESA WorldCover 2021	10 m	●
	Built-up fraction	ESA WorldCover 2021	10 m	
	Snow/ice fraction	ESA WorldCover 2021	10 m	
	Water fraction	ESA WorldCover 2021	10 m	
Climate	Bio1 (annual mean temp.)	CHELSAch	25 m	●
	Bio4 (temp. seasonality)	CHELSAch	25 m	●
	Bio12 (annual precip.)	CHELSAch	25 m	●
	Bio15 (precip. seasonality)	CHELSAch	25 m	
	Bio18 (precip. warmest quarter)	CHELSAch	25 m	
Biological	Canopy height	ETH Global Canopy Height	10 m	●
	NDVI (annual max)	GIMMS MODIS	250 m	
	Human Footprint	Mu et al. (2022)	1 km	●
	Forest-edge distance	Derived from tree cover	30 m	

Biological and anthropogenic features. Canopy height was obtained from the ETH Global Canopy Height Model at 10 m resolution (Lang et al., 2023) and regridded to 30 m via bilinear resampling. The annual maximum NDVI was computed from GIMMS MODIS 8-day composites (250 m, growing season DOY 97–273) and regridded to 30 m. The global Human Footprint index (Mu et al., 2022) at 1 km was bilinearly downsampled. Forest-edge distance was computed as a signed Euclidean distance transform from the tree-cover fraction layer (threshold 0.5): positive values indicate distance from forest, negative values indicate depth within forest.

2.3 Species Occurrence Data

Species occurrence records for vascular plants in Switzerland were obtained from the Global Biodiversity Information Facility (GBIF) open data snapshot hosted on Amazon S3 (GBIF.org, 2024). The raw dataset was filtered to retain only records within the study extent that met the following quality criteria: coordinate uncertainty ≤ 1000 m, valid geographic coordinates, and taxonomic rank at species level.

After filtering, the dataset comprised approximately 14 million occurrence records spanning 7,912 vascular plant species. For modelling purposes, we retained only species with at least 100 unique occurrence records, yielding 3,756 qualifying species. This threshold ensures sufficient training data for both the Random Forest baseline and the GNN-SDM, while acknowledging that it excludes many rare and threatened species with sparse observational records.

2.4 Protected Areas

Protected area boundaries for Switzerland were obtained from the World Database on Protected Areas (WDPA), filtered to nationally designated sites within the study extent. The dataset

includes national parks, nature reserves, landscape protection areas, and Ramsar sites. These boundaries are used in the gap analysis (??) to assess the degree to which high-priority habitat identified by the GNN-SDM falls within existing protected areas.

2.5 Feature Selection

To reduce multicollinearity and computational cost, we performed feature selection using Random Forest importance scores. A patch-level RF classifier was trained for six representative species spanning different habitat types and prevalence levels. Feature importance was computed as the mean decrease in Gini impurity, averaged across species.

From the initial 21 candidate features, 12 were retained for downstream modelling (Table 1). The excluded features were either highly correlated with retained variables (e.g., Bio15 with Bio12; plan curvature with profile curvature) or contributed negligible predictive power (e.g., NDVI at 250 m resolution added little beyond the 10 m canopy height and tree-cover fraction). The selected feature set balances ecological interpretability with predictive performance while keeping the dimensionality manageable for the Self-Organising Map clustering step.

3 Methods

This section describes the methodological pipeline for constructing a graph-based species distribution model (SOM-GNN-SDM) for Swiss flora at 30 m resolution, following the framework proposed by Wu et al. (2025). The pipeline consists of three stages: landscape patch construction via Self-Organising Map clustering and connected-component analysis, (2) a Random Forest baseline, and (3) the SOM-GNN-SDM itself.

3.1 Landscape Patch Construction

The landscape patch construction transforms the continuous 30 m raster feature stack into a graph representation where nodes correspond to ecologically homogeneous habitat patches and edges encode spatial adjacency.

3.1.1 SOM Clustering

A Self-Organising Map (SOM; Kohonen, 1982) was trained to identify recurring landscape types from the 12 selected environmental features. The SOM was configured as a 64×64 hexagonal, toroidal grid and initialised with PCA. Training used a subsample of 500 000 pixels (drawn from the ~ 160 M valid pixels in the study extent), preprocessed with a `RobustScaler` and domain-specific post-scaling (land-cover fractions up-weighted $3\times$; forest-edge distance clipped to $[-5, 5]$).

The SOM was trained for 150 epochs with a neighbourhood radius decaying from 64 to 16 and a learning rate decaying from 0.05 to 0.016. Convergence was assessed on a held-out 20% validation set using quantisation error (QE) and topographic error (TE); 150 epochs represented the sweet spot beyond which improvements were marginal.

After training, the codebook vectors were hierarchically clustered using Ward’s linkage, and the dendrogram was cut to yield 30 landscape types. Every valid pixel in the study extent was then assigned to its best-matching unit (BMU) via a FAISS exact- L_2 index, producing a categorical landscape-type raster at 30 m resolution.

3.1.2 Connected Components and Patch Merging

Spatially contiguous regions of the same landscape type were identified using connected-component labelling (`skimage.measure.label`). To avoid an excessive number of tiny fragments, patches

smaller than 100 pixels ($\sim 0.09 \text{ km}^2$) were merged into their most similar neighbour using a mapping-table approach. This procedure yielded **166,464 landscape patches** covering the full study extent.

For each patch, the mean value of the 12 selected environmental features was computed from the underlying raster stack, producing a $166,464 \times 12$ node-feature matrix.

3.1.3 Patch Network

An undirected graph was constructed by connecting patches that share at least one boundary pixel. The resulting **NetworkX** graph contains **166,464 nodes** and **465,833 edges**. Node attributes are the mean environmental features; no edge weights were used.

3.2 Baseline: Random Forest

A per-species Random Forest (Breiman, 2001) classifier served as the non-spatial baseline. For each of the 3,756 qualifying species (≥ 100 GBIF records), the following procedure was applied:

1. **Presence labels.** GBIF occurrence coordinates were mapped to the patch-label raster, yielding a set of presence patches per species.
2. **Pseudo-absence generation.** Background patches were sampled at a 3:1 ratio (absence : presence) from patches without recorded occurrences, using a fixed random seed for reproducibility.
3. **Model training.** A **scikit-learn** Random Forest was trained on the patch-level mean features using tuned hyperparameters (see below).
4. **Evaluation.** Performance was assessed via 5-fold cross-validated metrics: AUC, accuracy, precision, recall, F1, MCC, Cohen’s κ , and TSS.

Hyperparameter tuning was performed via **GridSearchCV** on the same six representative species used for the GNN architecture search, covering diverse habitat types across Switzerland. The best hyperparameters from this limited tuning were then applied uniformly to all 3,756 species.

3.3 GNN-SDM

3.3.1 Model Architecture

The SOM-GNN-SDM is based on GraphSAGE (Hamilton et al., 2017), implemented in PyTorch Geometric (Fey and Lenssen, 2019). At each layer l , a node v aggregates information from its neighbourhood $\mathcal{N}(v)$ using the mean aggregator and combines it with its own representation:

$$\mathbf{h}_{\mathcal{N}(v)}^{(l)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l-1)} \quad (1)$$

$$\mathbf{h}_v^{(l)} = \sigma(\mathbf{W}^{(l)} \cdot \text{CONCAT}(\mathbf{h}_v^{(l-1)}, \mathbf{h}_{\mathcal{N}(v)}^{(l)})) \quad (2)$$

where $\mathbf{h}_v^{(0)} = \mathbf{x}_v$ is the input feature vector of node v , $\mathbf{W}^{(l)}$ is a learnable weight matrix, and σ is the LeakyReLU activation function. Dropout ($p = 0.2$) is applied after each layer during training. The final node embeddings are passed through a linear layer with sigmoid activation to produce a habitat-suitability score $\hat{y}_v \in [0, 1]$ for every patch:

$$\hat{y}_v = \text{sigmoid}(\mathbf{w}^\top \mathbf{h}_v^{(L)} + b) \quad (3)$$

where L is the number of GraphSAGE layers.

An architecture search was conducted on six representative species spanning diverse habitats (*Picea abies*, *Fagus sylvatica*, *Trifolium pratense*, *Potentilla erecta*, *Phragmites australis*, *Senecio inaequidens*). Six architectures were compared:

- 1-layer: [16] (417 parameters)
- 2-layer: [24, 12] (1,201 parameters)
- Paper default: [24, 18, 8] (1,787 parameters)
- Wider: [48, 32, 16] (5,361 parameters)
- Extra-wide: [64, 48, 32] (10,929 parameters)
- 4-layer: [32, 24, 16, 8] (3,417 parameters)

The extra-wide 3-layer architecture [64, 48, 32] consistently achieved the highest mean AUC across all test species and was selected for production training. This is notably wider than the [24, 18, 8] architecture proposed by Wu et al. (2025) for virtual fauna species, suggesting that flora—with their sessile nature and finer-grained habitat associations—benefit from greater representational capacity per layer rather than additional depth.

3.3.2 Training Procedure

For each species, the GNN-SDM was trained independently using the following pipeline:

1. **Background selection via random pseudo-absence.** Non-presence patches were sampled at a 3:1 ratio (background:presence) using a fixed random seed. An initial design based on One-Class SVM background selection (as in Wu et al., 2025) was evaluated during development but abandoned in favour of random pseudo-absence for the production run. The random strategy ensures a fair comparison with the RF baseline (which uses same pseudo-absence protocol) and proved more robust across the full species set, avoiding the computational overhead and occasional instability of fitting a kernel SVM on high-dimensional patch features for each of 3,756 species.
2. **PageRank weighting.** Presence patches were weighted by their PageRank centrality in the landscape graph, following Wu et al. (2025). This assigns higher importance to structurally well-connected patches, reflecting their greater ecological significance as habitat corridors or core areas. Background patches received uniform weight of 1.0.
3. **Train/validation split.** The labelled patches (presence + background) were shuffled and split 80/20 into training and validation sets.
4. **Optimisation.** The model was trained with the Adam optimiser (learning rate = 0.001) using a weighted mean-squared-error loss:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} w_i (\hat{y}_i - y_i)^2$$

where $\mathcal{T}$ is the training set, w_i the PageRank-derived weight, $\hat{y}_i$ the predicted suitability, and $y_i \in \{0, 1\}$ the label.

5. **Early stopping.** Validation AUC was evaluated every 10 epochs. Training was halted if no improvement occurred for 50 consecutive epochs (patience). The model state with the best validation AUC was restored for prediction. Maximum training duration was 500 epochs.
6. **Prediction.** The best model was used to predict habitat suitability for all 166,464 patches, yielding a continuous suitability surface per species.

3.3.3 Evaluation

To evaluate the GNN-SDM on the same footing as the RF baseline, a post-hoc evaluation was performed for each species: presence patches and a 3:1 random sample of non-presence patches were combined, and the predicted suitability scores were evaluated using the same eight metrics (AUC, accuracy, precision, recall, F1, MCC, Cohen’s κ , TSS) with a classification threshold of 0.5.

3.3.4 Computational Setup

Production training of all 3,756 species was executed as an AWS SageMaker Training Job on a GPU instance (NVIDIA GPU, CUDA-accelerated). The job supported checkpointing for spot-instance interruptions and completed in approximately 50 minutes of GPU time. Node features were standardised (zero mean, unit variance) before training, and the full graph (166,464 nodes, 931,666 directed edges) was loaded onto the GPU for efficient message passing.

4 Results

4.1 Random Forest Baseline

TODO: RF results.

4.2 GNN-SDM Performance

TODO: GNN results and comparison.

4.3 Common vs. Vulnerable Species

TODO: Species-level analysis.

4.4 Habitat Suitability Maps

TODO: Map figures.

5 Discussion

5.1 Architecture Adaptation for Flora

TODO: Architecture discussion.

5.2 Value of Graph Structure

TODO: Graph structure value.

5.3 Limitations

TODO: Limitations.

5.4 Future Work

TODO: Future directions.

6 Conclusion

TODO: Write conclusion.

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TODO: Acknowledgements.

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